

Can a LoRA adapter leak its training images?

Yes — the gradient-bridge attack, in pictures. (MNIST odd/even, 784-1000-1000-1 MLP)

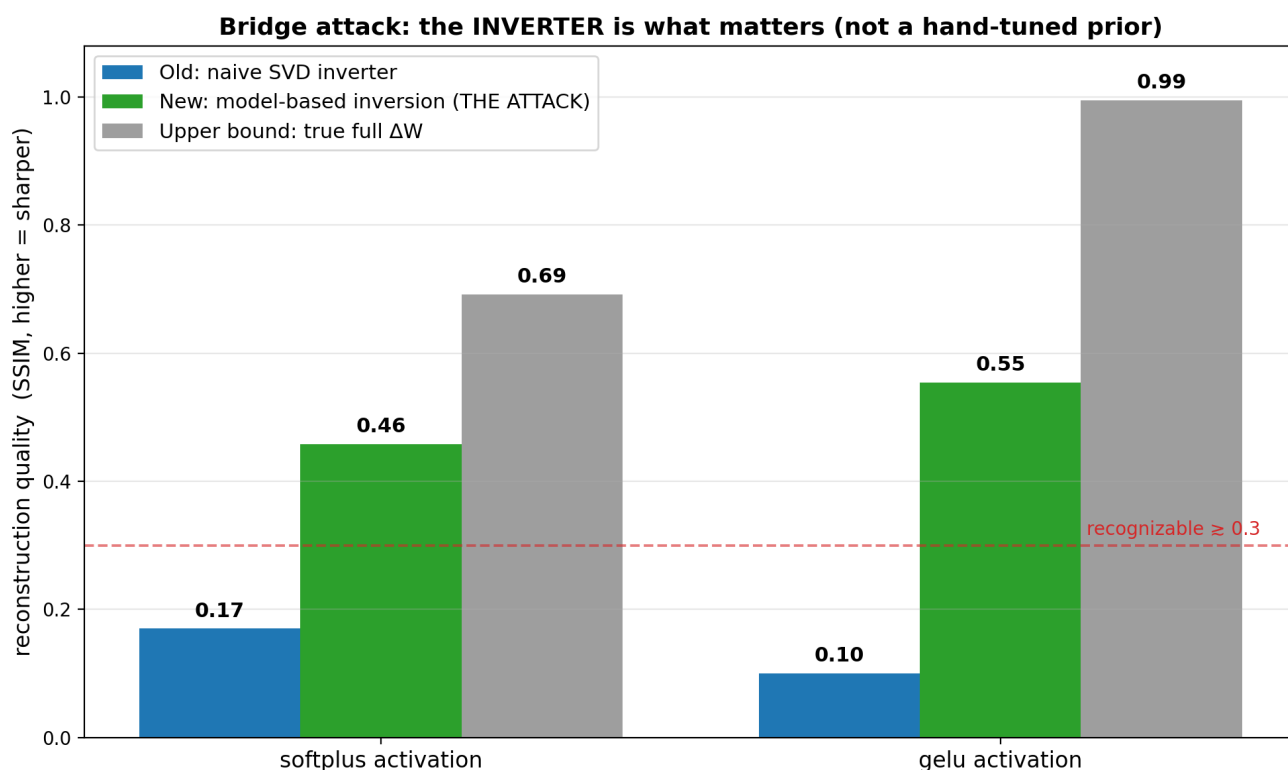
The setup. A model is fine-tuned on 2 private digits using a LoRA adapter. The attacker sees ONLY the tiny adapter — not the images, not the full weights. A small 'decoder' network, trained purely on PUBLIC MNIST, learns to turn the adapter back into the training gradient. That gradient is then inverted into an image.

The one finding. Getting a high-quality *gradient* is not enough. The step that turns the gradient into a picture — the **inverter** — is what decides success. A naive math inverter (SVD) gives a blur; inverting *through the known base model* gives a recognizable digit. Same gradient, 3-5x better image. This matches what we showed Gal before: the model/prior is the lever, not the raw similarity score.

How to read the image pages: each reconstruction grid has one column per private digit. The **top row is the real digit**; each row below is a reconstruction from a different setup. The row labelled **DECODED all-layers** is the real attack (adapter only, public-trained decoder). SSIM is printed on the left; 1.0 = identical, ≥ 0.3 = recognizable.

① The scoreboard (which inverter wins)

Reconstruction quality for each activation. Blue = the OLD naive-SVD inverter (a blur). Green = the NEW model-based inversion — the real attack — which clears the 'recognizable' line. Grey = the theoretical best (if you had the full weight change, not just the adapter). Experiment: job 445780, end-to-end.

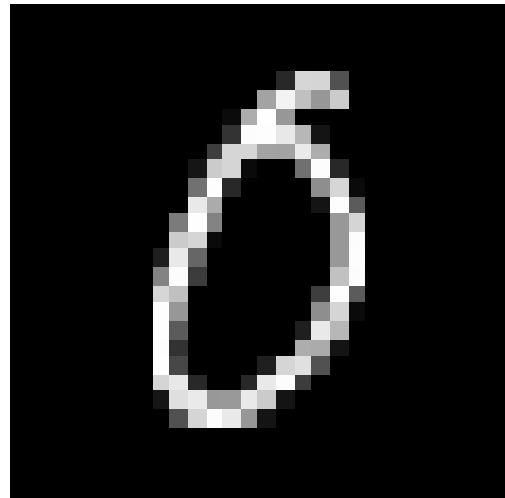
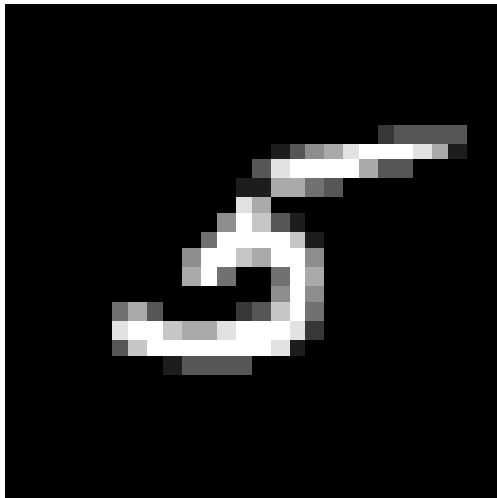


② THE ATTACK — reconstructions (softplus activation)

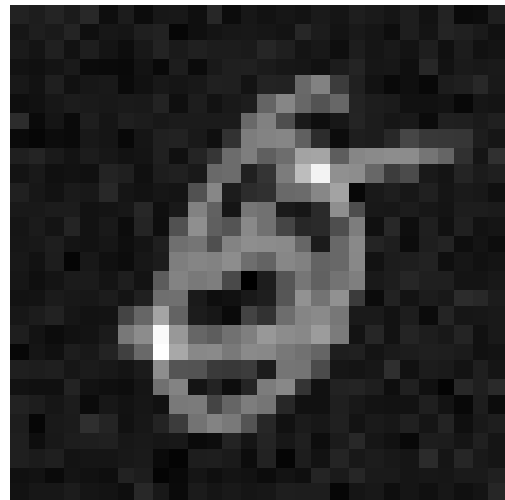
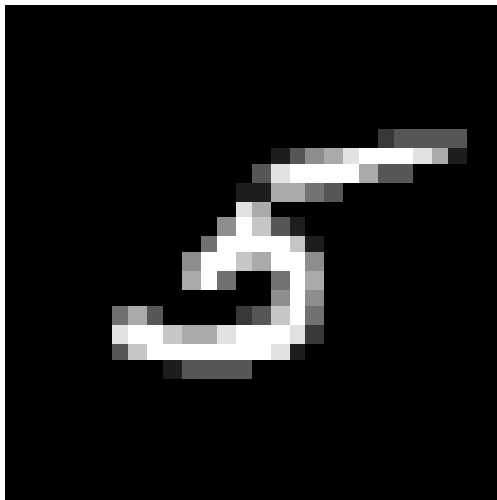
Experiment: end-to-end bridge (experiments/gradient_bridge/phase2_e2e.py). Top row = the true private digits. 'DECODED all-layers' and 'DECODED input-only' = the real attack (adapter only). 'TRUE ...' rows use the real weight change and are the best-case reference. The decoded rows are clearly the right digits, a bit soft — SSIM ~ 0.46 - 0.48 .

to-end (softplus): decoded $\Delta W \rightarrow$ model-

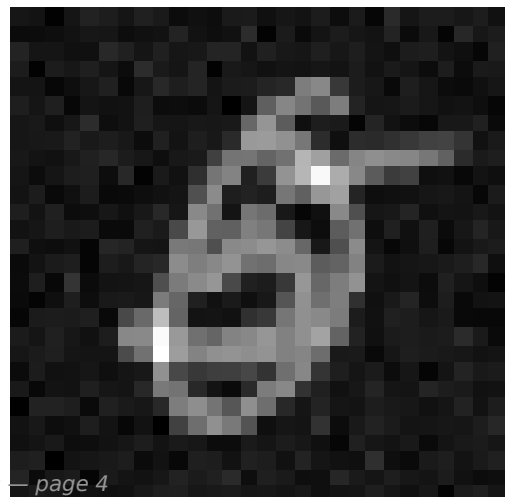
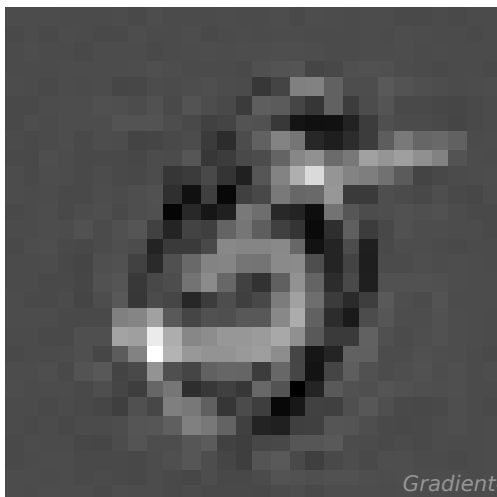
true



TRUE ΔW (ceiling)
ssim 0.69



DECODED all-layers
ssim 0.45

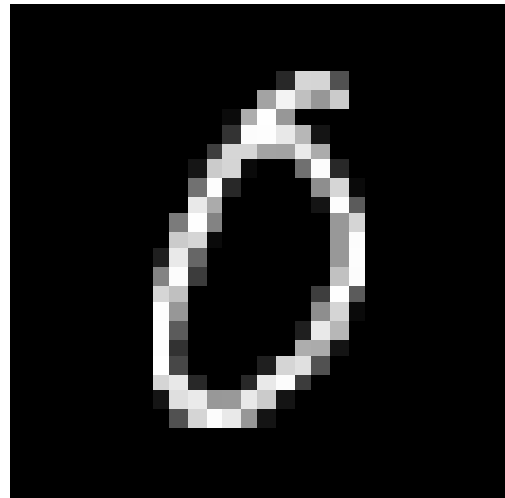
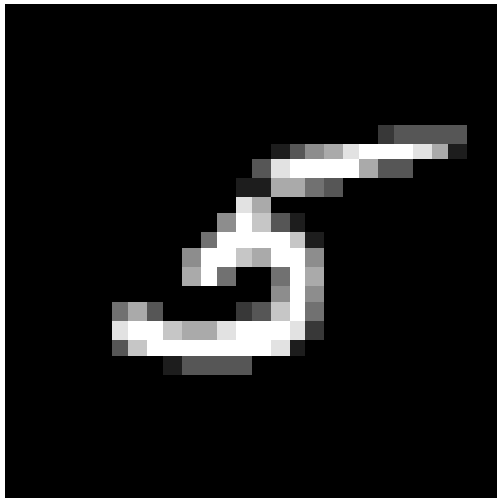


③ THE ATTACK — reconstructions (GELU activation)

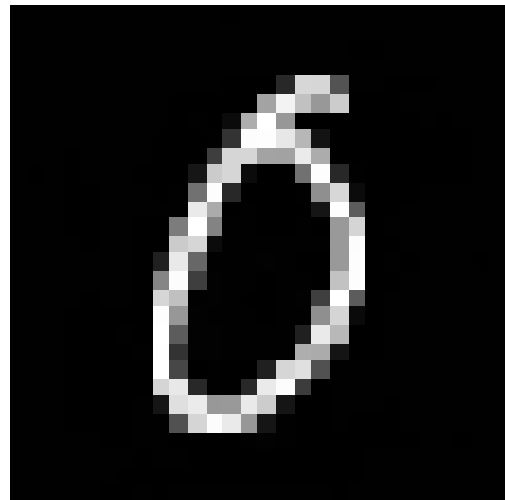
Same experiment, GELU activation (the deployment-realistic smooth activation). Here the true-weight reference is near-perfect (SSIM 1.0) and the decoded attack reaches SSIM ~ 0.55 — the sharpest bridge reconstruction we get.

d-to-end (gelu): decoded $\Delta W \rightarrow$ model-bi

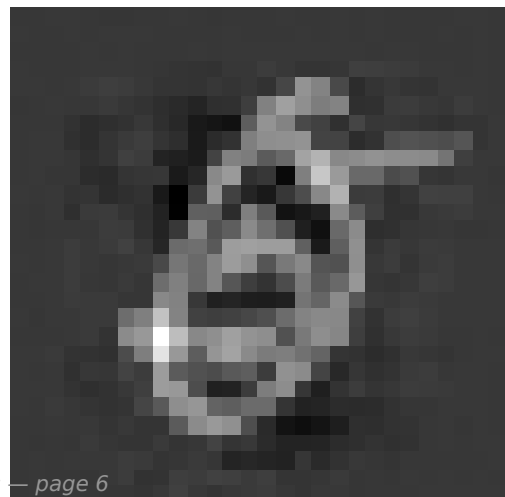
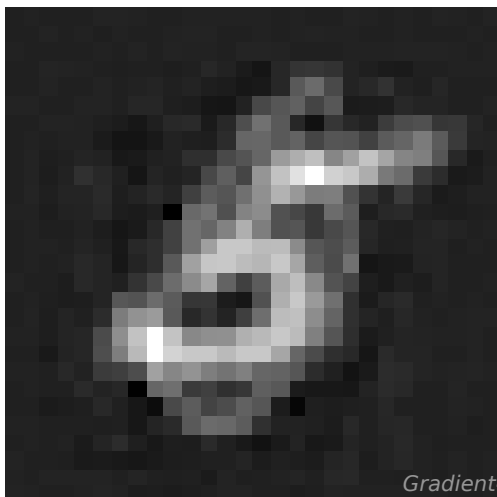
true



TRUE ΔW (ceiling)
ssim 1.00



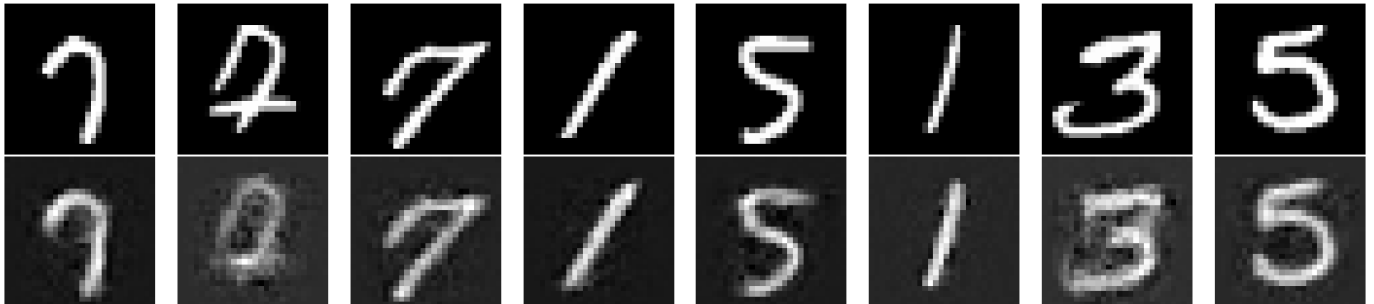
DECODED all-layers
ssim 0.56



④ Why the inverter matters — the WRONG one (naive SVD)

Experiment: input-layer SVD inversion (phase2_image.py). This takes the SAME decoded gradient as above but inverts it with plain linear algebra (SVD) instead of the model. Result: a coarse blob, SSIM ~ 0.17 . This is the corner the attack was stuck in before we switched inverters. Compare row-for-row with pages ②/③.

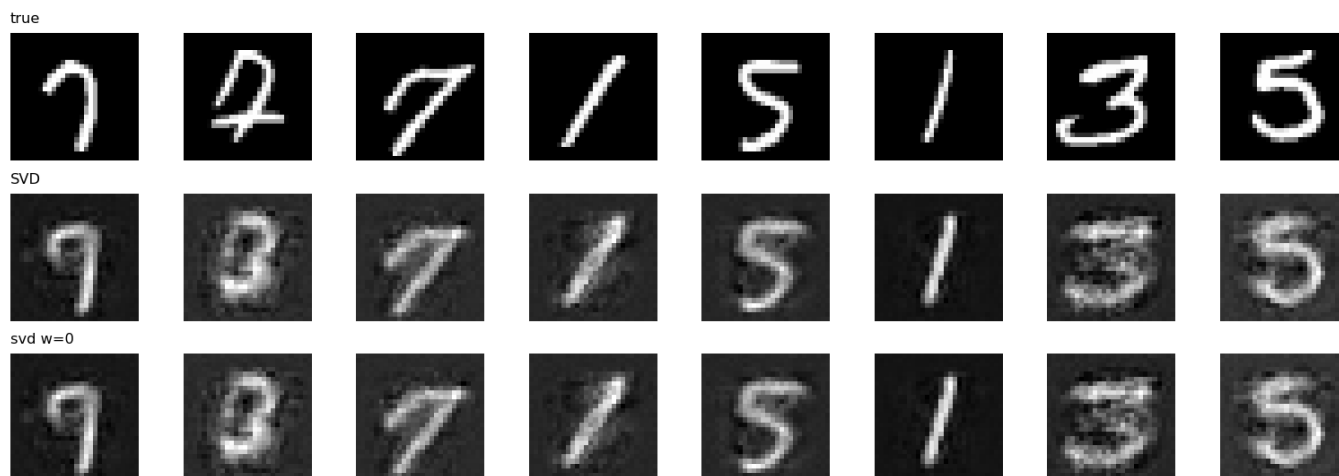
GB-Phase 2 (softplus): decoded-gradient -> image (top=true, bottom=recovered)



⑤ A dead end we ruled out — hand-crafted priors

Experiment: prior sweep (phase2_prior.py). We tried forcing the SVD image to be smooth (TV) or sparse (L1) — the usual tricks. Left col = truth, middle = raw SVD, right = best prior. Every prior made it WORSE, not better (see the STATUS note). That is what told us the problem was the inverter, not a missing prior — leading to the model-based fix on pages ②/③.

GB-Phase 2 priors (softplus): true / raw-SVD / best-prior



⑥ Under the hood — the decoder recovers the gradient well

Experiment: gradient-bridge decoder training (train_decoder.py). The decoder learns to rebuild the full-layer gradient from the tiny LoRA adapter. The curve climbs to ~ 0.99 cosine on held-out public data (this is the hidden layer). So the gradient recovery is strong — the remaining challenge is purely the inversion-to-pixels step.

